Machine Learning approach for sex classification using voice signal

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***Abstract*—This study evaluates the effectiveness of Particle Swarm Optimization (PSO) for feature selection in a gender classification task using Mel-Frequency Cepstral Coefficients (MFCCs) extracted from isolated vowel signals. A total of 185 samples were analyzed, and PSO was employed to identify the most discriminative subset of features for classification. Two machine learning models were trained, Random Forest Classifier (RFC) and Support Vector Machine (SVM) using only features selected by PSO. The algorithm consistently identified three MFCC coefficients (11, 12, and 14), which resulted in high classification accuracy for both models. The RFC achieved a test accuracy of 0.9459, while the SVM reached 0.9189. Statistical analysis revealed variance heterogeneity across the selected features; therefore, Welch's t-tests were applied, confirming significant differences between male and female groups. These findings demonstrate that PSO effectively reduces feature dimensionality while preserving classification performance. The results highlight the relevance of mid-to-high MFCCs for vowel-based gender recognition and support the use of evolutionary optimization approaches for acoustic feature selection in speech processing tasks.**

Keywords—Speech, Vowels, MFCC, SVM, RFC, Voice Recognition.

I. INTRODUCTION

Building a classification system that can recognize an individual's sex is complex. It requires the identification of features from voice signals capable of differentiating from subject to subject, while also finding the characteristics that represent the 'common factor' between a sex group [1].

Most of the time, the word 'gender' is used as a synonym for 'sex'. A person's gender can be studied through two approaches: essentialist and constructivist. In the constructivist vision, gender is described as a social construction and is associated with certain expectations. Most of the articles in the literature which we use for this report employ the essentialist approach. However, in this article, we will refer to the commonly used binary gender or gender as the real meaning, sex [2].

Identifying sex through the voice has been a challenging task in the research areas of speech processing, because human speech encompasses unique information such as age, emotional state, language accent, sex, tone, which conveys linguistic and non-linguistic features [3]. The lungs, trachea, vocal cords, larynx, nasal cavity, the lips, and the tongue are important elements that compose the speech production system; to produce the speech correctly, it's important that the starting organs and the various articulators are in correct coordination, the human brain is the center of that control. Expelling the air initiates the process, and the pressurized airs interacting with the vocal cord's vibration produces the air pulp. These vibrations make the vocal tone, which differs from person to person with aging, and fall in different ranges for males and females. These tones are modulated by other articulators, i.e. lips, tongue, teeth, the roof of the mouth, etc., to help produce various sounds by differentiating the way of their articulation [4]. In general, the voice is produced by a glottal excitation that is then filtered by the vocal tract to yield the air flow at the mouth, which is converted into a pressure waveform at the lips and propagated as sound waves [5].

Speech is the vocalized form of human communication; it is a non-stationary random signal. Based on its characteristics, it is classified into two categories: speech and silence, and the speech section is classified into voiced and unvoiced. The voiced part is the result of vowel utterance, whereas the unvoiced part is the outcome of the consonant utterance or due to the articulator's constrictions while speaking. It is called a quasi-periodic signal because it is a combination of the periodic vowel utterance and the non-periodic consonant utterance. The periodic vowels last for a longer duration, while the unvoiced exist for a shorter duration during a given period of the speech signal; hence, the voiced part has most of the energy of the signal [4]. That is the reason why we opted to use continuous vowel utterance to retrieve the audio samples to collect the necessary information to train our model.

The quality of the features selected to train the deep learning models, which depend on training sets, determines the effectiveness and robustness of the classifier. The efficiency of these classifiers can be improved by extracting proper voice features. In the literature, numerous methods have been developed and tested to extract feature sets from speech signals

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[3]. Many sex identification systems already exist; some of the literature mostly uses the pitch alongside the formant frequencies as key features to develop their classifiers, as it is seen in [4]; we opted to use the Mel Frequency Cepstral Coefficients (MFCC) to extract the relevant features, or coefficients, to classify the sex of the speaker as male or female from the voice samples obtained.

MFCCs represent a spectrum that models the vocal tube of humans, and the length of that tube is different from male to female [3]. Part of the process to extract the MFCCs is to transform the signal into the frequency domain using the Fourier transform (FFT), that so because the FFT is a vital signal processing tool that transforms the time domain into the frequency domain [4].

In this research, we aspired to correctly identify the sex of the speaker, extracting the features of the audio samples using MFCCs to determine them. To achieve this, we trained 2 different models, a Random Forest Classifier and a Support Vector Machine model, and at the end, we compared them to establish which of them gives the most accurate classification.

II. METHODOLOGY

A. Data obtention

For the data obtention, several recording sessions where conducted, the process to record each individual can be reviewed at following:

1. Introduce the subject to the measurement, explaining the goal of the session, without mentioning the classification objectives to avoid bias.
2. Prepare the recording device, bring the microphone close to the subject's mouth, approximately 3 cm from the mouth.
3. Count down from 3 and start the recording, then ask the person to start saying the corresponding vocal.
4. Repeat steps 2 and 3 for each vowel.

B. Samples characteristics

The total number of people recorded were 38, 22 men and 15 women. The age group varies from 17 to 23, having population in college phase. For each individual, 5 audios were taken, 1 for each vocal, giving a total of 185 audios that will be used for the classification task.

C. Recording device and sample rate

The voice recording process was conducted using 3 different devices, two having a sampling rate of 48 000 kHz, and other of 41 100 kHz. Because of this, we have adjusted the preprocessing of the audio in order to dismiss any variability that can be caused because of this.

D. Bandpass filter

To clean the audio samples, a bandpass filter was used. To select the minimum and maximum limits for the required band, the lower limit from the literature [5], which is 80 Hz for the lowest male vocal frequency, and the upper limit was determined from the maximum frequency registered from the audio samples, which is 5000 Hz.

E. Obtention of Cepstral Coefficients.

The MFCC, abbreviation for Mel Frequency Cepstral Coefficients, is an effective procedure often used to detect and identify features from voice signals [7]. The Cepstral Coefficients provide better prediction accuracy, compared to other parameters [8], therefore, for the main goal of this paper, these provide an excellent base for classification. As part of the Coefficients computing we included the log energy, with this, the expected output would be this value followed by the 13 coefficients. The methodology to extract these characteristics is defined by Figure 1.

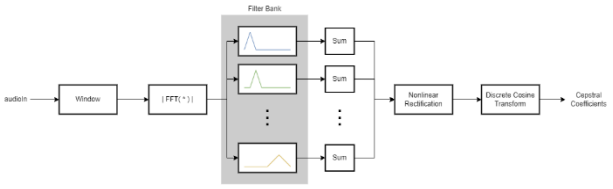


Fig. 1. Methodology to obtain the Cepstral Coefficients using Matlab's MFCC function [9].

As shown by the mentioned figure, the process starts by taking as input the audio signal, then a process named "Framing" occurs, in which the signal is being divided into small segments around the 20ms mark. With all the windows defined, an FFT is applied, this helps to identify the spectrum of the voice signals and is a vital step in acquiring the Coefficients [7]. Finally, a Discrete Cosine Transform (DCT) is used to convert the signal again into the time domain.

With this process, the Cepstral Coefficients are obtained; these valuable components of the signal will be later used for a more accurate prediction.

F. Dataset description

Based on the MFCC calculated from the samples, Table 1 & 2 (appendix A) shows descriptive data of all the features, where 0 is for female and 1 for male.

G. Data Normalization

Data normalization is essential to scale the numeric values of the features into a specified range without changing their nature; it is usually converted into the range [-3,3] to ensure that every variable has the same handling in the model and to enhance the training time. The data has been normalized by using the Eq 1:

$$z = \frac{x-\mu}{\sigma} \qquad (1)$$

Where x is the original value of the feature, μ represents the mean of the value of the feature, and σ is the standard deviation of the feature x.

H. Particle Swarn Optimization (PSO)

PSO is a popular optimization method inspired by the behavior of birds, either the search for food or a nest, and the interaction among their population during flight is modeled as an optimization mechanism [10].

The initial step of the PSO algorithm involves randomly selecting a set of particles, then changing the state of the particles to the optimum state based on their memory and knowledge to reach the best state. There are two main parameters for each particle: position and velocity, which are updated for the whole particles.

In each interaction, the particles have a position and a velocity in space. Each particle has a memory which stores its best achieved position (p_B) and the best position of the other particles known by it (g_B), namely, the best particle in a neighborhood. When the algorithm begins running, the position and velocity of the particles are randomly assigned. The Equations 2 and 3 refresh these parameters:

$$v_i(t+1) = v_i(t) + \psi_1\big(p_B - x_i(t)\big) + \psi_2(g_B - x_i(t)) \quad (2)$$

$$x_i(t+1) = x_i(t) + v_i(t+1), \quad (3)$$

where ψ_1 and ψ_2 are constants limited to an infinite interval [11]. In practice, the constants ψ_1 and ψ_2, which control the cognitive and social components of the particle's movement, are constrained within the range [0,1] to ensure stable convergence [12]. Similarly, in our implementation, the particle positions are defined within [0,1], representing the probability of including each feature in the subset.

In this study, PSO was applied for feature selection from the extracted MFFCs to improve the performance of two classifiers: Support Vector Machine (SVM) and Random Forest Classifier (RFC). Each particle was encoded as a binary vector representing the inclusion or exclusion of the feature. The objective function defined for the PSO algorithm is a fitness function that balances the accuracy of the model and the number of features used [13]. The fitness function is defined by equation 4:

$$cost = \alpha(1 - acc) + (1 - \alpha)\left(\frac{n_{features}}{n_{\max features}}\right), \quad (4)$$

Where acc refers to the accuracy of the classifier, $n_{features}$ is the size of feature subset, and $n_{\max features}$ represents the original feature size. α is constant, which has values between 0 and 1 to define the weights of the goal [13]; for our study, it was set to 0.9.

The PSO algorithm iteratively updates the particles' positions and velocities, guiding the swarm toward feature subsets that optimize classification accuracy and number of features. This approach allows for efficient exploration of the feature space, reduces overfitting, and enhances model generalization.

I. Machine Learning Model

Now, after selecting the best Cepstral Coefficients, a Machine Learning (ML) algorithm can be used to classify a subject birth sex, before doing so, we explored some of the best options supported by the literature [7][14], those will be reviewed in the posterior sections.

J. Train and test dataset

First, we divided the signal samples into the train/test datasets, this is done to ensure the model is being created only using a determined set of data, so it can be tested afterwards using the corresponding test dataset. This practice is vital in the ML process and guarantees no data leaks occur during the training, meaning that the model has not 'seen' any possible samples that may be used to compute accuracy and other performance metrics.

The approach we followed is to divide the set of data using an 80/20 relationship for the train/test dataset, stratifying each of them using the sex variable. This is due the unbalance of classes that exists; this step reduces the possibility of variations or underperforming in the 'women' class.

K. Cross Validation

While training a ML model, overfitting may occur sometimes, in which the model adjusts more than necessary to the training data, and this causes the model to underperform when trying to predict for samples it has not seen yet [15]. In order to prevent this situation, Cross Validation (CV) is implemented.

The CV process can be executed by different approaches, one of which is the K-fold CV. This works by dividing the training set in 'K' small groups of data, then, iterations have place in which all the folds except for one are used to train the model, with the model trained the accuracy is measured, and the process repeats for 'K' iterations, changing the fold that is employed for the testing, as seen in Fig. 3.

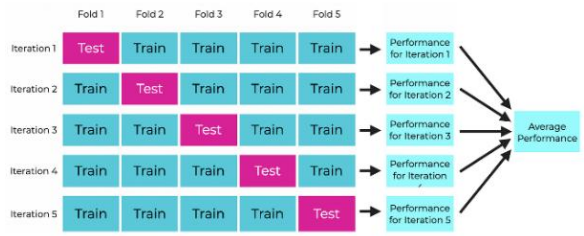


Fig. 3. Example of a K-fold Cross Validation [16]

L. Support Vector Machines

Support Vector Machines (SVM) is a ML technique that is commonly used to speech recognition tasks [16], thanks to its versatility and ease of adaptation to the investigation necessities.

This algorithm works by creating a hyperplane, as shown by fig. 4, this, in a general description, is similar to a vector that intends to divide or separate correctly two or more groups of data. This works by recognizing characteristics of the variables, so the algorithm can distinguish between them. An advantage provided by this approach is that the Vector used to differentiate subjects does not depend on a 2D arrangement, it can scale to more dimensions as needed, although the interpretability of model will diminish as its complexity arises [17].

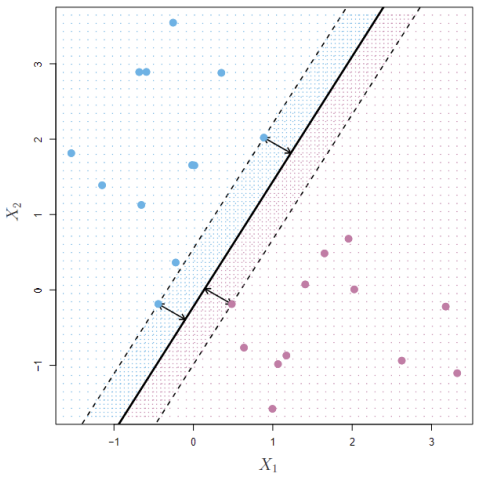


Fig. 4. Example of a linear SVM model dividing two classes [17].

The kernel used for the SMV model is the Radial Basis Function (RBF), a kernel, for this context, can be understand as the function used to create the hyperplane, and the RBF works by creating a circular-like plane that divides both classes.

M. Random Forest

A decision tree is an easy-to-understand classification algorithm commonly used because of its interpretability and compatibility with the human approach to determine whether an outcome is from one type or another. This is because the 'tree' is like making a series of decisions, in certain order, that ends with only one possible classification for a sample. [17]

Although this seems like a great option for the main goal of the investigation, a more robust approach is needed to ensure that the results are reliable. This is the reason a Random Forest (RF) is preferred, this variant of a tree-classification algorithm is, in base, a set of 'trees' put together to try and classify the sample with their correct 'tag'. [17]

Random Forests take its name from having trees that get trained using different predictors each time a tree is trained, this forces the model to use and consider every possible 'route' to make the prediction and diminish the correlation between them. [17]

Finally, after all the trees have been trained, and their decisions are compared, the most common decision made would become the classification for the sample, making this approach robust and capable of performing a sex recognition task.

For the RF model, 100 decision trees were implemented, having balance between having a computational-efficient model, but keeping sufficient performance while preventing overfitting.

III. RESULTS

To evaluate the effectiveness of the proposed feature selection approach using PSO we trained and tested two classifiers—SVM and RFC—on the extracted MFCC features. The performance was assessed in terms of classification accuracy on the test set.

The initial analysis focused on the subset of features selected by PSO and its impact on model performance. Table 1 summarizes the selected features.

Feature Extraction using PSO			
Classifier	***# Selected Features***	***Selected features IDs***	***Accuracy (Test)***
RFC	3	11, 12, 14	0.94
SVM	3	11, 12, 14	0.89

Table 1. Feature Extraction using PSO algorithm.

For subsequent analysis, tests for normalization were made in selected features by PSO algorithm divided by gender using Shapiro-Wilk test. Every feature shows a normal distribution; Table 2 summarizes the results for each feature:

Normality Test for Selected Features				
		Mean	***Std. Deviation***	***P-value of Shapiro-Wilk***
Coeff11	0	0.108	0.228	0.81
Coeff11	1	-0.179	0.417	0.094

Coeff12	0	-0.095	0.244	0.548
Coeff12	1	0.218	0.349	0.399
Coeff14	0	0.214	0.166	0.6
Coeff14	1	-0.157	0.283	0.249

Table 2. Normality test for selected features by PSO algorithm using Shapiro-Wilk test.

We evaluated the equality of variances within the features selected by the PSO. Table 3 presents the results, showing that none of the variables meets the assumptions of the Levene's test.

Test of Equality of Variances (Levene's)				
	F	***df₁***	***df₂***	***p***
Coeff11	21.954	1	183	5.434×10^{-6}
Coeff12	9.347	1	183	0.003
Coeff14	13.986	1	183	2.462×10^{-4}

Table 3. Test of Equality of Variances for selected features.

Consequently, we performed an independent samples t-test (Welch's test) to determine whether the groups exhibit differences in the MFCC features. Table 4 presents the Welch test results.

Independent Samples T-Test (Welch's test)			
	t	***df***	***p***
Coeff11	6.02	175.905	9.892×10^{-9}
Coeff12	-7.169	182.866	1.810×10^{-11}
Coeff14	11.22	179.337	1.799×10^{-22}

Table 4. Independent Samples T-Test for selected MFCCs.

To further assess the effectiveness of the PSO-based feature selection strategy, both classifiers (RFC and SVM) were retrained using only the three selected MFCC features (IDs 11, 12, and 14). Unlike the fitness function used during PSO optimization—which included a penalty term to avoid excessive feature usage—model performance in this stage was evaluated using standard classification metrics on the test set.

Table 5 summarizes the performance of both classifiers.

Performance of RFC and SVM models on Test Set			
Classifier	***Accuracy***	***Recall***	***F1-Score***
RFC	**0.9459**	**0.9545**	**0.9545**
SVM	0.9189	0.9091	0.9302

Table 5. Comparison of performance of classifiers.

In general, RFC model had a better performance than SVM. For better visualization, Figure 5 & 6 shows both confusions matrixes of the classifiers. For better comprehension refer to Appendix B.

At the individual prediction level, both models show nearly identical performance. A sample-wise comparison revealed that only one instance was classified differently between the SVM and the RFC.

IV. Discussion

The results show that only three MFCC features (11, 12, and 14) were sufficient to achieve high classification accuracy in both models. Because the analysis was conducted using isolated vowel signals, these MFCCs likely capture formant-related spectral differences between male and female voices. This suggests that PSO effectively identified the most discriminative acoustic information for gender classification in vowel-based speech signals.

Interestingly, the selected MFCCs correspond to higher-order cepstral coefficients, which capture subtler spectral variations rather than general energy or pitch. This aligns with previous findings showing that mid-to-high MFCC indices often encode formant-related differences between male and female voices. PSO proved effective for dimensionality reduction, consistently selecting the same three MFCC features for both classifiers. This stability suggests that the optimization converged towards a robust subset of discriminative features, reducing overfitting risk and improving model interpretability.

The lack of homogeneity of variance revealed by Levene's test suggests that male and female MFCC distributions have different variabilities, which is not a negative point, its most well expected because of the frequency and other speech features differ from one person to another, even though they are similar between one man to another and one woman to another. The significant Welch's test results for the selected features further confirm the existence of acoustic differences between groups, supporting the classifier's ability to separate them.

While analyzing model's performance, it is noted that RFC outperformed the SVM, achieving 0.9459 accuracy compared to 0.9189. This may be attributed to RFC's ability to capture non-linear relationships and interaction between MFCC features, which are commonly present in speech signals. Although the RF model has better prediction power, as shown by the accuracy and other metrics, the computational cost of it

is greater than the SVM approach, making it less reliable to implement as a solution in devices with limited performance, and, because of this, SVM should not be discarded as a reliable option for sex identification.

Both models performed great in other vital metrics such as recall, having 0.95 and 0.91 for RCF and SVM respectively. This guarantees both algorithms have low false negatives detection, which is helpful for applications that involves interaction with persons, and for which, the detection of their sex can be used for other purposes, such as illness diagnosis, in which this variable has a direct impact on an individual health.

A limitation of this study is the relatively small sample size, which may affect generalizability. Additionally, sample size between male and female were unbalanced, possibly affecting the general performance of the model, enhancing the models to perform in favor of male category.

Moreover, only MFCC features were considered; incorporating prosodic features such as pitch or jitter may further improve performance.

V. Conclusion

During this research, different techniques were studied to properly construct a classifier that could identify the sex of the speaker using audio samples. To achieve this, two models were trained using MFCCs as key features: the RFC and the SVM. This study demonstrates that Particle Swarm Optimization (PSO) is an effective feature selection method for gender classification using MCC features from isolated vowel signals. PSO consistently identified a compact subset of three MFCC coefficients (11, 12, and 14), which proved sufficient to achieve high test accuracy in both classifiers evaluated. The Random Forest Classifier (RFC) achieved 0.9459 accuracy, while the Support Vector Machine (SVM) reached 0.9189 accuracy using the same selected features.

The results indicate that mid-to-high-order MFCCs capture discriminative spectral differences between male and female vowel productions, allowing robust performance even when most features are removed. Additionally, statistical analyses revealed that the selected features did not meet variance homogeneity assumptions; therefore, Welch's t-test was applied, confirming significant group differences in these MFCCs.

Overall, the findings suggest that PSO not only reduces the dimensionality but also enhances the model interpretability by identifying the most acoustically relevant features for vowel-based gender classification. This approach may be extended in future work to continuous speech, larger datasets, and additional classifiers to validate its generalization capabilities.

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Appendix A

	MFCC1		***MFCC2***		***MFCC3***		***MFCC4***		***MFCC5***		***MFCC6***		***MFCC7***	
Gender	0	1	0	1	0	1	0	1	0	1	0	1	0	1
Mean	0.203	1.659	-14.71	-13.23	3.555	3.948	0.236	0.266	1.415	1.775	-0.87	-0.64	0.593	0.287
Std. Deviation	2.421	2.05	2.835	2.696	1.092	1.069	0.998	0.898	0.927	1.048	0.442	0.643	0.429	0.378

Table 1. Descriptive data of MFCCs (1-7).

	MFCC8		***MFCC9***		***MFCC10***		***MFCC11***		***MFCC12***		***MFCC13***		***MFCC14***	
Gender	0	1	0	1	0	1	0	1	0	1	0	1	0	1
Mean	-0.15	-0.02	0.202	0.084	-0.20	8.87×10^{-4}	0.172	-0.08	-0.14	0.142	-0.08	-0.12	0.186	-0.20
Std. Deviation	0.213	0.427	0.259	0.296	0.34	0.318	0.212	0.407	0.226	0.331	0.233	0.325	0.16	0.262

Table 2. Descriptive data of MFCCs (8-14).

Appendix B

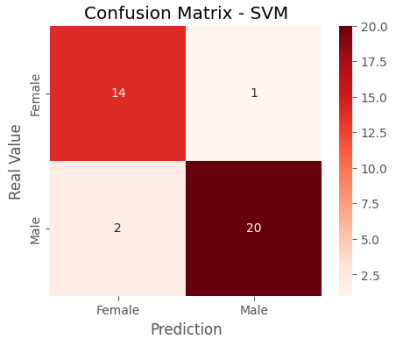


Fig. 5. Confusion matrix using SVM model.

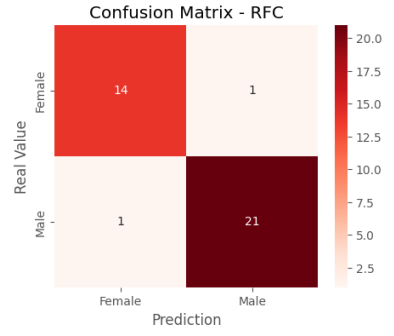


Fig. 6. Confusion matrix using RFC model.

Honor code

We hereby affirm that we have done this activity with academic integrity.